

Differential Privacy against Membership Inference Attacks

A Case Study on Binary Classifiers

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Presentation Outline

01 Membership Inference Attacks

Definition, Threat and Threat Model

02 Target Model Implementation

Vulnerable binary classifier on the Adult dataset

03 Shadow Model Attack

Attack architecture and pipeline

04 Differential Privacy (DP-SGD)

Mechanism, Opacus and privacy budget ϵ

05 Iterative Tuning & Model Collapse

Balancing issues and adopted solutions

06 Results & Conclusions

Utility-privacy trade-off and final metrics



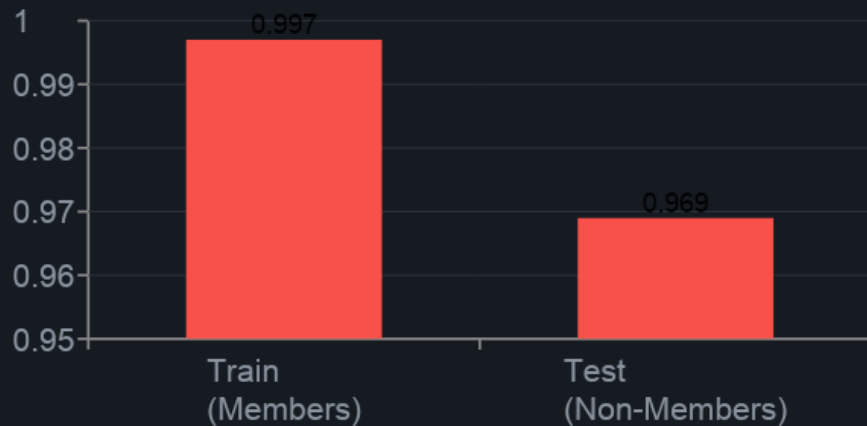
What is a Membership Inference Attack?

Definition: An attacker, given a data record x and access to the model (black-box or white-box), tries to determine whether $x \in D_{\text{train}}$.

Why is it dangerous?

- ML models may memorize patterns from the training data
- In MLaaS settings, the attacker can freely query the model
- Training records may reveal sensitive information, such as income or health data

Signal exploited by the attack



Adult Dataset & Vulnerable Target Model

Adult Income Dataset

Task: binary classification ($>50K$ / $\leq 50K$)

Features: 14 attributes (age, occupation, education...)

Target split: 500 samples (250 train / 250 test)

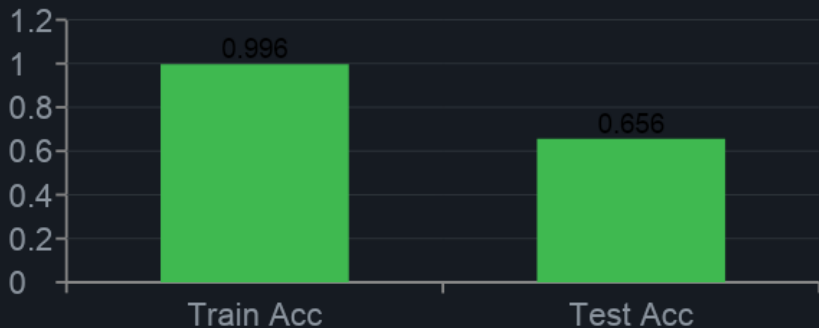
Target MLP Architecture (epochs 200)

Linear(Input_Size, 256) $\rightarrow$ ReLU $\rightarrow$

Linear(256, 128) $\rightarrow$ ReLU $\rightarrow$

Linear(128, 64) $\rightarrow$ ReLU $\rightarrow$ Linear(64, 2)

Intentional Overfitting Gap



Baseline Metrics (no DP)

Train Accuracy	99.6%
Test Accuracy	65.6%
Overfitting Gap	34.0%

Attack Types

Black-Box Score-Based

- Access only to predictions
- Observes confidence vectors
- Exploits the train/test gap
- Used in this work
- Approach: shadow models

FOCUS OF THIS WORK

Model-Agnostic (ML-Leaks)

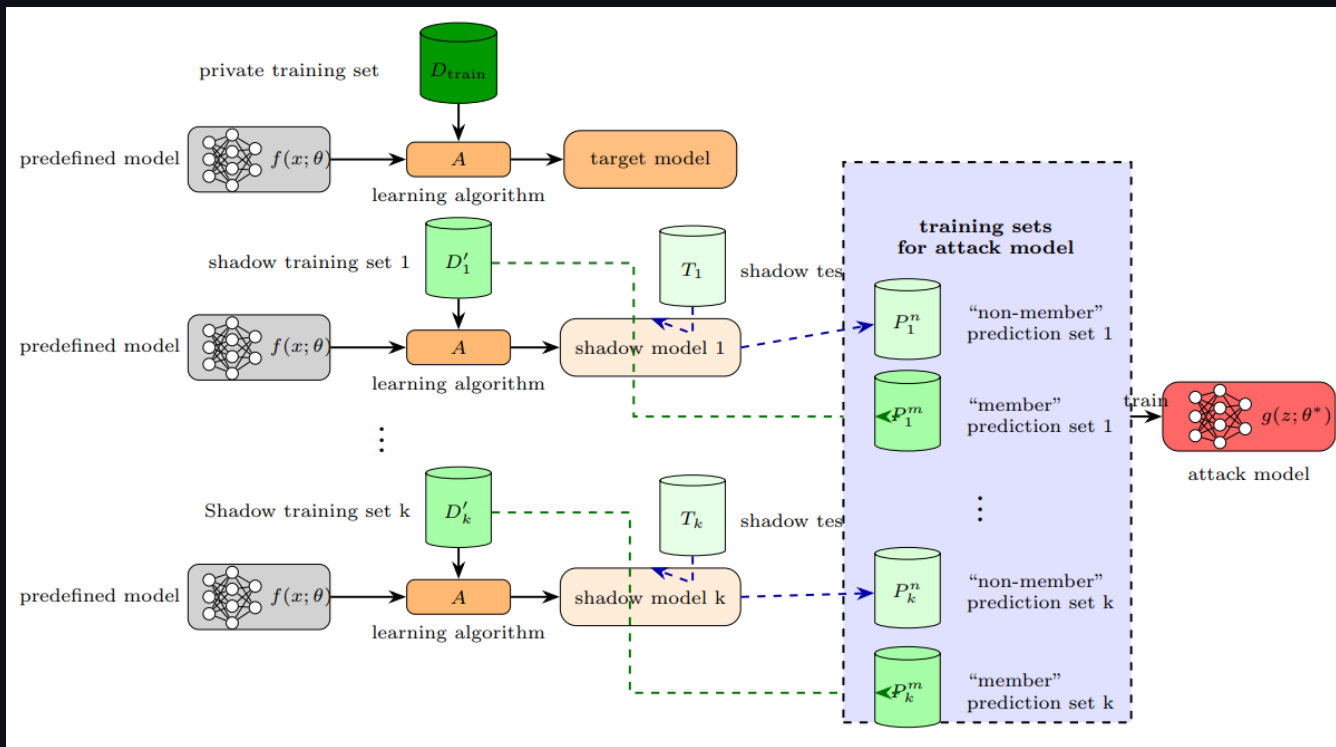
- Does not require an identical architecture
- Uses only the score distribution
- Salem et al. 2019
- More generalizable
- No overlap between dataset

FOCUS OF THIS WORK

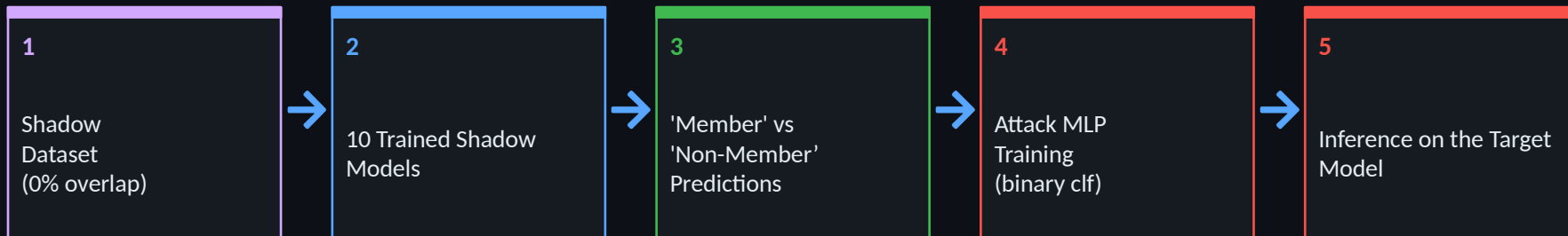
White-Box Gradient

- Access to weights and gradients
- Uses influence functions
- Leino & Fredrikson 2020
- More powerful
- Not covered in this study

Shadow Models



Attack Pipeline with Shadow Models



Based on: Shokri et al. 2017 · ML-Leaks: Salem et al. 2019

Implementation Details

Component

Shadow Models

Shadow MLP

Attack Classifier

Baseline Result

Configuration

10 (with the same distribution as the target)

Different layers from the Target Model (ML-Leaks generalization)

Binary MLP: [In/Out] from the vector confidence

Attack Accuracy 68.6% · Member Recall 1.00 · F1 0.76

Shadows & Attack model

Shadows architecture (epochs 100)

`Linear(Input_Size, 128) → ReLU →`

`Linear(128, 64) → ReLU → Linear(64, 2)`

Attack model architecture (epochs 100)

`Linear(Number_of_classes + 2, 64) → ReLU →`

`Dropout(0.3) → Linear(64, 32) → ReLU →`

`Linear(32, 2)`

```
# Max confidence
train_max_conf = train_probs.max(axis=1).reshape(-1, 1)
test_max_conf = test_probs.max(axis=1).reshape(-1, 1)
```


Vulnerability of the Unprotected Model

99.6%

Train Accuracy

65.6%

Test Accuracy

34.0%

Overfitting
Gap

68.6%

Attack
Accuracy

Confidence Gap = 0.0281



Attack Metrics



Theoretical Foundations of DP-SGD

Definition of (ϵ, δ) -DP: A mechanism M is (ϵ, δ) -differentially private if, for every pair of adjacent datasets D and D' and every subset S of the output space: $Pr[M(D) \in S] \leq e^{\epsilon} \cdot Pr[M(D') \in S] + \delta$

ϵ (Epsilon)

Privacy Budget

Measures the maximum privacy loss.
Smaller values imply stronger privacy.

δ (Delta)

Failure Probability

Probability that the privacy bound fails.
Dynamically set as: $\delta = 1/|X_{\text{train}}| = 0.004$

σ (Sigma)

Noise Multiplier

Controls the intensity of the Gaussian noise injected into the gradients.

References: Abadi et al. 2016 – Deep Learning with Differential Privacy (CCS) · Humphries et al. 2020 – DP does not bound MIA in practice

Opacus: Meta's Library for DP in PyTorch

```
privacy_engine = PrivacyEngine()  
model, optimizer, train_loader  
    = privacy_engine.make_private(  
    module=model,  
    optimizer=optimizer,  
    data_loader=train_loader,  
    noise_multiplier=1.5,  
    max_grad_norm=1.5  
    )
```

Privacy tracking in the code:

```
epsilon = privacy_engine.get_epsilon(delta)
```

Target threshold: $\epsilon < 10$ · Final result: $\epsilon \approx 5.31$

1

Per-Sample Gradients

Gradients computed for each individual sample, not immediately averaged

2

Gradient Clipping

L2 norm clipped to $\text{max_grad_norm}=1.5$ to limit the influence of outliers

3

Noise Addition

Calibrated Gaussian noise ($\sigma=1.5$) added to the clipped gradients

4

Privacy Accounting

Opacus tracks the accumulated ϵ after epoch

Model Collapse: the trap of illusory privacy



First DP attempt (clipping=1.0, σ =1.0, 200 epochs, batch=16): $\epsilon = 29.35$, Test Accuracy $\approx 75.6\%$... but the confusion matrix showed **class 1 with F1-score of exactly 0.00**.

Class Distribution (Training set)



Adopted correction path

1

Class-Weighted Loss

Only partial improvement, sampling rate $q=0.256$ was too high

2

Oversampling

Prevented collapse but duplicated data, with a risk of spurious overfitting

3

Balanced Stratified Subset ✓

250 samples per class, NO artificial duplication → final solution

Search for the Optimal DP Configuration

Test	Batch	σ (Noise)	Epochs	Test Acc	Train Acc	Gap	ϵ
1 ★	32	1.5	40	69.6%	72.4%	2.8%	8.43
2	32	1.3	40	70.0%	72.0%	2.0%	10.54
3	32	1.5	30	67.2%	70.8%	3.6%	7.18
4	16	1.3	40	68.0%	68.8%	0.8%	6.98

Test 1 was the best result of the initial search, but not the final DP configuration. The final setup was obtained after further tuning: `max_grad_norm = 1.5` and `$\delta = 1 / |X_{train}|$` , instead of a fixed default δ value designed for larger datasets. This led to the final privacy budget $\epsilon \approx 5.31$.

The Utility-Privacy trade-off

↓ σ

Reducing σ increases accuracy but increases ϵ (weaker privacy)

↓ E

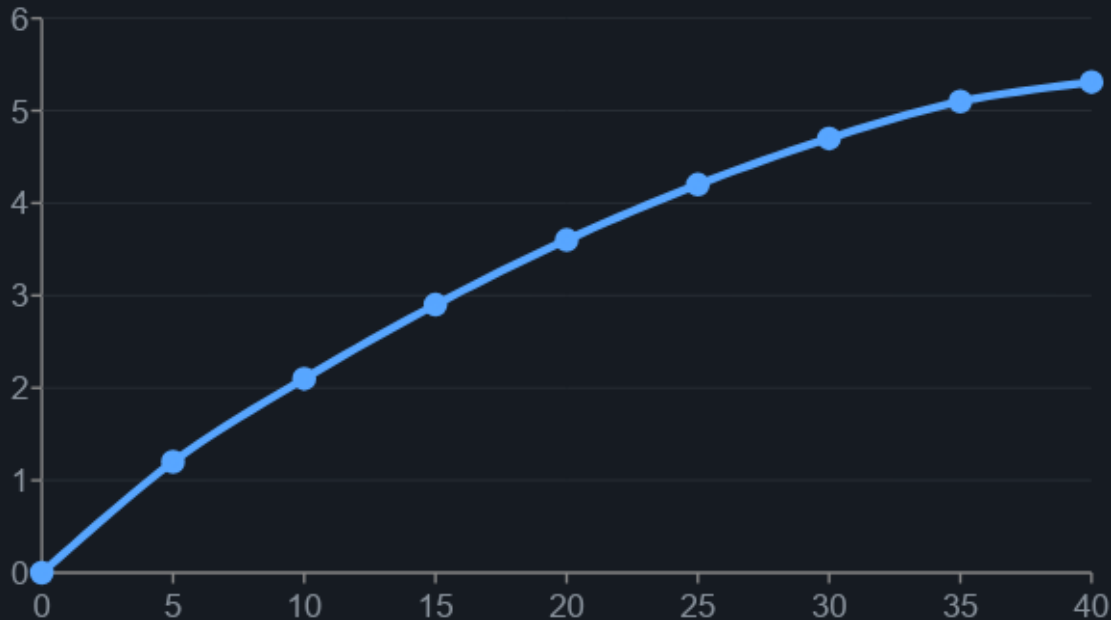
Reducing the number of epochs limits ϵ accumulation but decreases accuracy

↓ B

Reducing the batch size improves privacy but makes training less stable

Evolution of ϵ during DP training

Privacy Budget ϵ per Epoch



Commonly accepted safety threshold: $\epsilon < 10$

δ (failure prob.)

0.004

$1 / |X_{\text{train}}|$

σ (noise mult.)

1.5

final configuration

max_grad_norm

1.5

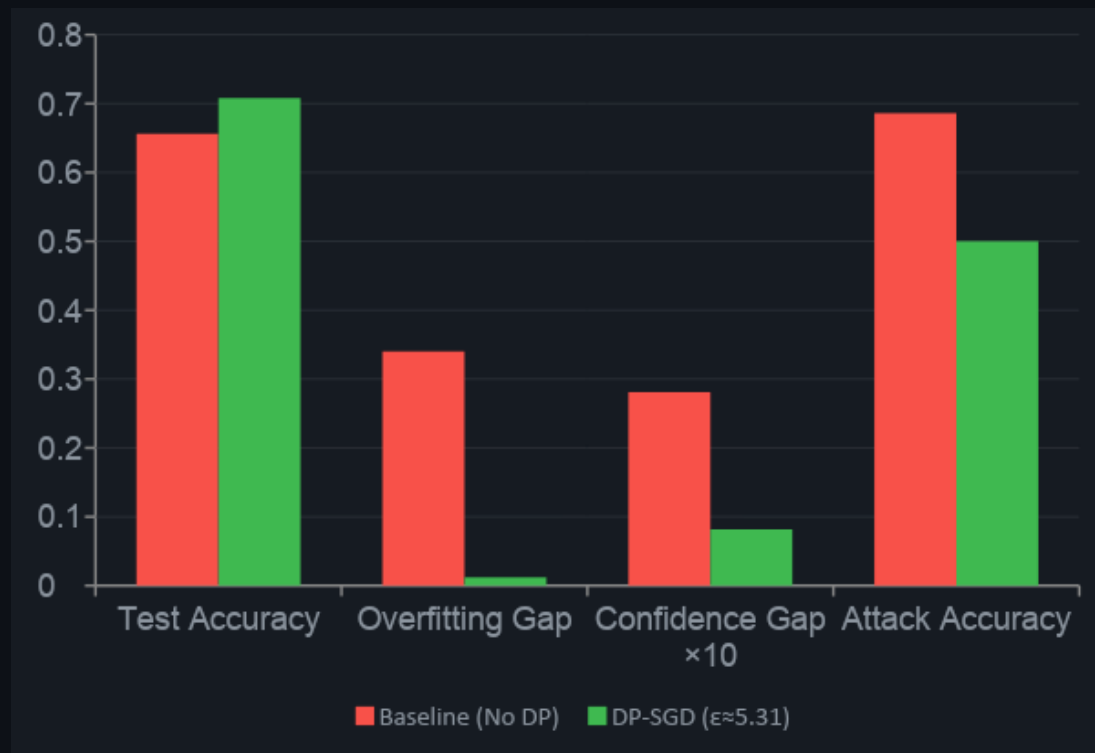
clipping threshold

final ϵ (40 epochs)

5.31

below the $\epsilon < 10$ threshold

Comparison: Baseline vs DP-SGD



Overfitting Gap

34.0%

→

1.2%**↓ 97%**

Confidence Gap

0.0281

→

0.0082**↓ 71%**

Attack Accuracy

68.6%

→

≈50%**↓ random**

Test Accuracy

65.6%

→

70.8%**(+5.2%)**

Critical Analysis of Results and Limitations



What Worked

- ✓ DP-SGD reduced the overfitting gap from 34.0% to 1.2%
- ✓ The confidence gap decreased from 0.0281 to 0.0082 (-71%)
- ✓ Attack accuracy dropped to $\approx 50\%$ (close to the random baseline)
- ✓ Test accuracy even improved: 65.6% $\rightarrow$ 70.8%
- ✓ The balanced dataset avoided illusory privacy
- ✓ $\epsilon \approx 5.31$ with $\delta = 0.004$: the formal privacy budget was respected



Limitations



Very small dataset ($n=500$): DP scales better on larger dataset



$\delta = 1/250$ accelerates budget depletion $\rightarrow$ more fragile tuning



Only black-box score-based attacks were covered; white-box attacks were not evaluated



Oversampling may introduce spurious patterns in privacy-sensitive contexts



Generalization to NLP or deep architectures requires further validation

Summary and Future Work

DP-SGD ($\epsilon \approx 5.31$) effectively suppressed membership inference signals by reducing the overfitting gap to **1.2%** and bringing attack accuracy down to **$\approx 50\%$ (random baseline)**, while maintaining acceptable utility (Test Accuracy: **70.8%**). The key methodological correction was the use of a **perfectly balanced dataset (50/50)** to avoid illusory privacy.

Future Work



ROC-AUC and advanced metrics

Integration of ROC-AUC curves to evaluate inference boundaries



White-Box Analysis

Extension to white-box attacks with gradient access (Leino & Fredrikson)



Comparison with Regularization

Benchmark DP-SGD vs weight decay, dropout, early stopping



Larger dataset

Validate DP on real-scale tabular dataset (MLaaS setting)

Thank you for your attention



Any Questions?

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